

EXERCISE 13 Creating Views

1. What are three uses for a view from a DBA's perspective?

1. Security: Restrict user access to specific columns or rows
2. Simplification: simplify complex queries for end users
3. Data Consistency: Present consistent, read only data to multiple users.

2. Create a simple view called view_d_songs that contains the ID, title and artist from the DJs on Demand table for each "New Age" type code. In the subquery, use the alias "Song Title" for the title column.

Create View view_d_songs As
 Select id, title As "Song Title", artist
 From djs-on-demand
 Where type-code = 'NewAge';

3. SELECT * FROM view_d_songs. What was returned?

Select * From view_d_songs displays:
 Displays id, Song title, and artist columns

4. REPLACE view_d_songs. Add type_code to the column list. Use aliases for all columns.

Create OR Replace View view_d_songs As
 Select id As "ID", title As "Song Title", type-code As "Type Code"
 From djs-on-demand
 Where type-code = 'NewAge';

Or use alias after the CREATE statement as shown.

5. Jason Tsang, the disk jockey for DJs on Demand, needs a list of the past events and those planned for the coming months so he can make arrangements for each event's equipment setup. As the company manager, you do not want him to have access to the price that clients paid for their events. Create a view for Jason to use that displays the name of the event, the event date, and the theme description. Use aliases for each column name.

```
Create View view_events - Jason AS  
Select event_name AS "Event Name",  
       event_date AS "Event Date",  
       theme_desc AS "Theme Description"  
From events;
```

6. It is company policy that only upper-level management be allowed access to individual employee salaries. The department managers, however, need to know the minimum, maximum, and average salaries, grouped by department. Use the Oracle database to prepare a view that displays the needed information for department managers.

```
Create View dept_salary_summary AS  
SELECT department_id AS "Dept ID",  
       MIN(salary) AS "Minimum Salary",  
       MAX(salary) AS "Maximum Salary",  
       AVG(salary) AS "Average Salary"  
FROM employees  
GROUP BY department_id;
```

DML Operations and Views

Use the DESCRIBE statement to verify that you have tables named copy_d_songs, copy_d_events, copy_d_cds, and copy_d_clients in your schema. If you don't, write a query to create a copy of each.

1. Query the data dictionary USER_UPDATABLE_COLUMNS to make sure the columns in the base tables will allow UPDATE, INSERT, or DELETE. All table names in the data dictionary are stored in uppercase.

```
SELECT table_name, column_name, insertable, updatable, deletable
FROM user_updatable_columns
WHERE table_name = 'COPY_D_SONGS';
```

Use the same syntax but change table_name of the other tables.

2. Use the CREATE or REPLACE option to create a view of all the columns in the copy_d_songs table called view_copy_d_songs.

```
CREATE OR REPLACE VIEW view_copy_d_songs AS
SELECT * FROM copy_d_songs;
```

3. Use view_copy_d_songs to INSERT the following data into the underlying copy_d_songs table. Execute a SELECT * from copy_d_songs to verify your DML command. See the graphic.

ID	TITLE	DURATION	ARTIST	TYPE_CODE
88	Mello Jello	2	The What	4

```
INSERT INTO view_copy_d_songs (id, title, duration, artist,
                                type_code)
VALUES (88, 'mello Jello', 'The what', 4);
```

Verify :-

```
SELECT * FROM copy_d_songs;
```

4. Create a view based on the DJs on Demand COPY_D_CDS table. Name the view read_copy_d_cds. Select all columns to be included in the view. Add a WHERE clause to restrict the year to 2000. Add the WITH READ ONLY option.

```
CREATE OR REPLACE VIEW read_copy_d_cds AS
SELECT *
FROM copy_d_cds
WHERE year = 2000
WITH READ ONLY;
```

5. Using the read_copy_d_cds view, execute a DELETE FROM read_copy_d_cds WHERE cd_number = 90;

```
DELETE FROM read_copy_d_cds WHERE cd_number = 90;
```

6. Use REPLACE to modify read_copy_d_cds. Replace the READ ONLY option with WITH CHECK OPTION CONSTRAINT ck_read_copy_d_cds. Execute a SELECT * statement to verify that the view exists.

```
CREATE OR REPLACE VIEW read_copy_d_cds AS
SELECT *
FROM copy_d_cds
WHERE year = 2000
WITH CHECK OPTION CONSTRAINT ck_read_copy_d_cds;
```

7. Use the read_copy_d_cds view to delete any CD of year 2000 from the underlying copy_d_cds.

```
DELETE FROM read_copy_d_cds WHERE year = 2000;
```

8. Use the read_copy_d_cds view to delete cd_number 90 from the underlying copy_d_cds table.

```
DELETE FROM read_copy_d_cds WHERE cd_number = 90;
```

9. Use the read_copy_d_cds view to delete year 2001 records.

```
Delete from read_copy_d_cds where year = 2001;
```

10. Execute a SELECT * statement for the base table copy_d_cds. What rows were deleted?

SELECT * FROM copy_d_cds;

11. What are the restrictions on modifying data through a view?

A view cannot modify data if:
* it contains group by distinct functions
* uses join between multiple tables

12. What is Moore's Law? Do you consider that it will continue to apply indefinitely? Support your opinion with research from the internet.

* Moore law states that the number of transistors on a microchip doubles roughly every two years leading to exponential growth

13. What is the "singularity" in terms of computing?

The technological singularity is a theoretical point where artificial intelligence surpasses human intelligence leading to rapid, unpredictable technological growth

Managing Views

1. Create a view from the copy_d_songs table called view_copy_d_songs that includes only the title and artist. Execute a SELECT * statement to verify that the view exists.

```
Create view view_copy_d_songs as  
Select title, artist  
From copy_d_songs;
```

2. Issue a DROP view_copy_d_songs. Execute a SELECT * statement to verify that the view has been deleted.

```
DROP view view_copy_d_songs;  
SELECT * FROM user_views  
WHERE view_name = 'view-copy-d-songs';
```

3. Create a query that selects the last name and salary from the Oracle database. Rank the salaries from highest to lowest for the top three employees.

```
SELECT last_name, salary  
FROM employees  
ORDER BY salary Desc  
FETCH only FIRST 3 Rows ONLY;
```

4. Construct an inline view from the Oracle database that lists the last name, salary, department ID, and maximum salary for each department. Hint: One query will need to calculate maximum salary by department ID.

```
SELECT e.last_name, e.salary, e.dept_id, d.max_salary  
FROM employees e  
JOIN (SELECT dept_id, MAX(salary) AS max_salary  
FROM employees GROUP BY dept_id)
```

5. Create a query that will return the staff members of Global Fast Foods ranked by salary from lowest to highest.

```
SELECT staff_name, salary FROM global_fast_foods  
ORDER BY salary ASC;
```

Indexes and Synonyms

1. What is an index and what is it used for?

An index is a database object that improves the speed of data retrieval from a table. It works like a book index - allowing quick access without scanning the whole table.

2. What is a ROWID, and how is it used?

ROWID is a unique address of a row in a table it's used internally by Oracle to locate rows quickly.

3. When will an index be created automatically?

An index is automatically created when a column is declared as a primary key or unique constraint.

4. Create a nonunique index (foreign key) for the DJs on Demand column (cd_number) in the D_TRACK_LISTINGS table. Use the Oracle Application Express SQL Workshop Data Browser to confirm that the index was created.

Create INDEX dj_cd_idx

ON d_track_listings (cd_number);

To confirm

SELECT index_name, table_name,
FROM user_indexes

5. Use the join statement to display the indexes and uniqueness that exist in the data dictionary for the DJs on Demand D_SONGS table.

SELECT index_name, table_name, uniqueness

FROM user_indexes

WHERE table_name = 'D_SONGS';

6. Use a SELECT statement to display the index_name, table_name, and uniqueness from the data dictionary USER_INDEXES for the DJs on Demand D_EVENTS table.

SELECT index_name, table_name, uniqueness

FROM user_indexes

WHERE table_name = 'D_EVENTS';

7. Write a query to create a synonym called dj_tracks for the DJs on Demand d_track_listings table.

CREATE SYNONYM dj_tracks FOR d_track_listings;

8. Create a function-based index for the last_name column in DJs on Demand D_PARTNERS table that makes it possible not to have to capitalize the table name for searches. Write a SELECT statement that would use this index.

CREATE INDEX Partners-last-name-idx

ON d-Partners (UPPER(last_name));

SELECT * FROM d-partners

WHERE UPPER(last_name) = 'SMITH';

9. Create a synonym for the D_TRACK_LISTINGS table. Confirm that it has been created by querying the data dictionary.

```
CREATE SYNONYM dj-track FOR d-track-listings;  
SELECT synonym_name, table_name  
FROM user_synonyms WHERE synonym_name = 'D3_Tracks';
```

10. Drop the synonym that you created in question

DROP SYNONYM dj-track;

Evaluation Procedure	Marks awarded
Query(5)	5
Execution (5)	5
Viva(5)	5
Total (15)	15
Faculty Signature	<i>P. J. M.</i>